1️⃣ Manipulate numbers, graphical and statistical data (Item 7)

Relevance: ★★★★★ (Most critical)

Description:

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

Manifestation in the problem:

Students must process and calculate various numerical values, including:

- Area calculations
- Multiplication using price per square meter
- Addition of additional value components
- Final price comparison

Operations involved:

- Multiplication
- Addition
- Numerical comparison

Therefore, **numerical manipulation is the core competency in this task**.

2️⃣ Use a multi-step procedure leading to a mathematical solution (Item 12)

Relevance: ★★★★★

Description:

> Use a multi-step procedure leading to a mathematical solution.

The solution requires several steps:

1. Extract information from the table
2. Calculate the base price
3. Determine the appropriate additional value categories
4. Calculate the additional values
5. Compute the total estimated value
6. Compare it with the selling price
7. Draw a conclusion

Thus, the problem represents a **typical multi-step problem-solving task**.

3️⃣ Make sense of, relate and use a variety of representations (Item 10)

Relevance: ★★★★☆

Description:

> Make sense of, relate and use a variety of representations when interacting with a problem.

The problem includes multiple representations:

- Text description
- Property information table
- Evaluation criteria table

Students must translate between these representations:

Table information → numerical interpretation → mathematical calculations

Thus, **representation interpretation** is an important competency here.

4️⃣ Understand and utilise constructs based on definitions and rules (Item 5)

Relevance: ★★★★☆

Description:

> Understand and utilise constructs based on definitions, rules and formal systems.

The problem provides a **clear rule-based valuation system**, including:

- Price per square meter
- Additional values based on conditions

Students must understand and correctly apply these rules.

5️⃣ Use an understanding of the context to guide solving (Item 13)

Relevance: ★★★☆☆

Description:

> Use an understanding of the context to guide solving.

Students need to understand the real-world context of:

- Real estate valuation
- Apartment size
- Location value

This contextual understanding helps interpret the evaluation rules.

6️⃣ Articulate a solution (Item 8)

Relevance: ★★★☆☆

Description:

> Articulate a solution, showing and/or summarising intermediate mathematical results.

The question asks students to:

"Show that the selling price is very good."

Therefore, students must:

- Present the calculation process
- Explain the conclusion

1️⃣ Manipulate numbers, graphical and statistical data（第7项）

相关性：★★★★★（最核心）

能力描述：

> Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations.

在本题中的体现：

学生需要处理和运算多种数值信息：

- 面积计算
- 单价乘法
- 附加价值累加
- 最终价格比较

涉及运算：

- 乘法
- 加法
- 数值比较

因此 **数值操作能力是本题最核心能力。**

2️⃣ Use a multi-step procedure leading to a mathematical solution（第12项）

相关性：★★★★★

能力描述：

> Use a multi-step procedure leading to a mathematical solution.

解题过程需要多个步骤：

1. 从信息表中提取数据
2. 计算基础价格
3. 判断附加价值类别
4. 计算附加价格
5. 计算总估价
6. 与实际售价比较
7. 得出结论

因此属于 **典型多步骤问题求解。**

3️⃣ Make sense of, relate and use a variety of representations（第10项）

相关性：★★★★☆

能力描述：

> Make sense of, relate and use a variety of representations when interacting with a problem.

题目包含多种表征：

- 文本信息
- 房屋信息表
- 估价规则表

学生需要在这些表示之间进行转换：

表格信息 → 数值判断 → 数学计算

因此涉及 **多表征理解能力。**

4️⃣ Understand and utilise constructs based on definitions and rules（第5项）

相关性：★★★★☆

能力描述：

> Understand and utilise constructs based on definitions, rules and formal systems.

题目给出了一个**明确的估价规则系统**：

- 每平方米价格
- 不同条件下的附加价值

学生需要理解并正确应用这些规则。

5️⃣ Use an understanding of the context to guide solving（第13项）

相关性：★★★☆☆

能力描述：

> Use an understanding of the context to guide solving.

学生需要理解现实情境：

- 房地产估价
- 房屋面积
- 地理位置价值

理解这些背景有助于学生理解题目中的估价规则。

6️⃣ Articulate a solution（第8项）

相关性：★★★☆☆

能力描述：

> Articulate a solution, showing and/or summarising intermediate mathematical results.

题目要求：

“Show that the selling price is very good.”

因此学生需要：

- 展示计算过程
- 解释结论